

AWS EC2 Lab: Create an EC2 Instance and Assign a New Security Group

Step-by-step procedure with screenshots and explanations

1. Objective

Create a simple Amazon EC2 instance using the AWS Management Console, use a VPC and subnet, create a new security group, allow SSH access from the current public IP, and verify that the instance is running and the security group is attached.

2. Network configuration used

Component	Value / Purpose
VPC	Default VPC — 172.31.0.0/16
Subnet	My-EC2-Subnet — 172.31.0.0/20
Availability Zone	ap-south-1a
Public IP	Enabled
Security Group	My-EC2-SG
Inbound rule	SSH / TCP / Port 22 / My IP
AMI	Amazon Linux 2023
Instance type	t3.micro

3. Step-by-step procedure

Step 1 — Launch the EC2 instance

Open AWS Management Console → EC2 → Instances → Launch instances. Enter the instance name My-First-EC2.

It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices.

Take a walkthrough Do not show me this message again.

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.

Recents Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs
including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.18 AMI

ami-0a62b311c2f9b7a64 (64-bit (x86), x86_64) / ami-0cf34bdc963a29d7 (64-bit (x86), x86_64)

Description

Amazon Linux 2023 (kernel-6.18) is a modern, general purpose Linux-based OS that will be supported until June 2029. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.12.20260950.0 x86_64 HVM kernel-6.18

Architecture

64-bit (x86)

Boot mode

uEFI-preferred

AMI ID

ami-0a62b311c2f9b7a64

Creation Date

2023-09-29

Username

ec2-user

Verified provider

Instance type

t3.micro

Family: t3 2 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0112 USD per hour On-Demand SUSE base pricing: 0.0112 USD per hour On-Demand Ubuntu Pro base pricing: 0.0147 USD per hour On-Demand RHEL base pricing: 0.04 USD per hour

Free tier eligible

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

Create new key pair

Summary

Number of instances

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.12...read more

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

Explanation: The launch screen shows the selected Amazon Linux 2023 AMI, t3.micro instance type and key pair.

Step 2 — Select the VPC and subnet

In Network settings, select the default VPC (172.31.0.0/16), select My-EC2-Subnet (172.31.0.0/20), and keep Auto-assign public IP enabled.

Explanation: A VPC is the logical AWS network. A subnet is a smaller IP range inside that VPC. The subnet places the EC2 instance in a specific Availability Zone.

Step 3 — Create the new security group

Choose Create security group. Set the security group name to My-EC2-SG. Keep the SSH inbound rule on TCP port 22 and use My IP as the source.

Network settings

VPC - required | Info

vpc-089571e53df2467b4 (default) [Info](#)

Subnet | Info

subnet-057da4a92dfb5c01 My-EC2-Subnet [Info](#) [Create new subnet](#)

VPC: vpc-089571e53df2467b4 Owner: 741180449510 Availability Zone: ap-south-1a (aps1-az1) Zone type: Availability Zone IP addresses available: 65530 CIDR: 172.31.0.0/16

Auto-assign public IP | Info

Disable

Firewall (security groups) | Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

[Create security group](#) [Select existing security group](#)

Security group name - required

launch-wizard-1

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _; /!@,[]+=&()*~

Description - required | Info

launch-wizard-1 created 2026-10-02T04:16:55.555Z

Inbound Security Group Rules

▼ Security group rule 1 (TCP; 22, 122.171.23.104/32) [Remove](#)

Type	Protocol	Port range	Source type	Name	Description - optional
ssh	TCP	22	My IP	122.171.23.104/32	e.g. SSH for admin desktop

[Add security group rule](#)

► Advanced network configuration

Summary

Number of instances | Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.12...[read more](#)
ami-08c3b31556c937a94

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Explanation: A security group acts as a virtual firewall for the EC2 network interface. Port 22 is the standard SSH port for Linux remote administration. My IP restricts SSH access to the detected public IP.

Step 4 — Launch and verify the instance

Click Launch instance. Open the Instances page and wait until the instance state becomes Running and status checks pass.

EC2 > Instances

Dashboard
AWS Global View
Events

Instances

Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers
Capacity Manager

Images

AMIs
AMI Catalog

Elastic Block Store

Volumes
Snapshots
Lifecycle Manager

Network & Security

Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces

Instances (1/1)

Filter Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs	Monitoring	Security group name
My-First-EC2	i-09bc1c00112d2b352	Running	t3.micro	3/3 checks passed	ap-south-1a	ec2-13-233-198-164.ap...	13.233.198.164	-	-	disabled	My-EC2-SG

i-09bc1c00112d2b352 (My-First-EC2)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

Instance summary

Instance ID i-09bc1c00112d2b352	Public IPv4 address 13.233.198.164 open address	Private IPv4 addresses 172.31.141.140
IPv6 address -	Instance state Running	Public DNS ec2-13-233-198-164-ap-south-1.compute.amazonaws.com open address
Hostname type IP name: ip-172-31-141-140.ap-south-1.compute.internal	Private IP DNS name (IPv4 only) ip-172-31-141-140.ap-south-1.compute.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t3.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address 13.233.198.164 [Public IP]	VPC ID vpc-089571e53df2467b4	Auto Scaling Group name -
IAM role -	Subnet ID subnet-057da4a92dfb5c01 (My-EC2-Subnet)	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:ap-south-1:741180449510:instance/i-09bc1c00112d2b352	
Operator -		

Explanation: The screenshot confirms My-First-EC2 is Running, uses t3.micro, is in ap-south-1a, and has a public IPv4 address.

Step 5 — Verify the security group

Select My-First-EC2 → Security tab. Check that My-EC2-SG is attached and that the inbound rule shows port 22/TCP from the configured public IP.

EC2

Instances

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

UltraServers

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Instances (1/1)

Info

Connect

Instance state

Actions

Launch instances

Last updated 8 minutes ago

Filter Instance by attribute or tag (case-sensitive)

Choose filter set

Name

Instance ID

Instance state

Instance type

Status check

Availability Zone

Public IPv4 DNS

Public IPv4 ...

Elastic IP

IPv6 IPs

Monitorin

My-First-EC2

i-09bc1c00112d2b552

Running

t3.micro

3/3 checks pass

ap-south-1a

ec2-15-233-198-164.ap...

15.233.198.164

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disabled

i-09bc1c00112d2b552 (My-First-EC2)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Security details

IAM role

--

Owner ID

741180449510

Launch time

Fri Oct 02 2026 09:41:33 GMT+0530 (India Standard Time)

▼ Security groups

sg-08b5e29ff6a6e79f (My-EC2-SG)

▼ Inbound rules

Filter rules

Name

Security group rule ID

Port range

Protocol

Source

Security groups

Description

--

sgr-0368c01a3be605e7d

22

TCP

122.171.23.104/32

My-EC2-SG

--

Explanation: This is the final verification that the newly created security group is assigned to the EC2 instance and SSH access is allowed only from the specified source IP.

4. Final result

The EC2 instance My-First-EC2 was successfully created in the default VPC and My-EC2-Subnet. A new security group named My-EC2-SG was assigned to it. The instance is running with a public IPv4 address, and the security group allows SSH (TCP port 22) from the configured My IP source.

5. Important security and cost notes

- For SSH, prefer My IP rather than allowing 0.0.0.0/0 unless a lab specifically requires public SSH access.
- Keep the downloaded .pem private key secure and do not share it.
- When the lab is finished, terminate the EC2 instance if it is no longer needed to avoid unnecessary charges. Also review any other billable resources you created.